

What an 8-Bit Frame Can and Cannot Say About the Scene Behind It

Bounds for inverse tone mapping, and a gate that reaches one of them

Ahmed Abdelnaby

Ahmed Talaat

FXTD Studios, Cairo, Egypt

4 September 2026

Abstract

We train a compact network for single-image SDR-to-HDR reconstruction and evaluate it against its own analytic inverse tone map on 429 held-out frames in PU21-PSNR and ColorVideoVDP JOD. On degraded input — an unknown tone curve, 4:2:0 chroma, banding and JPEG, the condition real footage arrives in — the model gains **+1.43 dB and +0.44 JOD**, winning **348 of 429 frames (81%)**. On clean, well-graded input it *loses* **3.0 dB** of PU21-PSNR while CVVDP puts the same gap at **−0.046 JOD**: a two-order-of-magnitude disagreement between the metrics that is itself the result.

The clean regression is not uniform. It concentrates entirely in low-dynamic-range frames: the 60 worst average **−10.76 dB** at a median ground-truth peak of **238 nits**, the 60 best **+3.41 dB** at **19,590 nits**, with correlation **+0.46** between gain and $\log_2(\text{peak_nits})$. The model has no representation of how much headroom a frame has, and so cannot decide to do nothing. The penalty is carried by the tail of the error distribution rather than its bulk (§6), which is itself a caution about the metric.

We then measure what fixing that would be worth, and whether it can be fixed. An oracle per-frame scale on the residual is worth **+5.84 dB on clean and +0.29 dB on hard simultaneously**, and no constant achieves it — clean wants 0.125, hard wants 1.1, and the constant that fixes clean discards 85% of the hard gain. But a linear readout of the features such a gate would use explains only **3%** of that scale’s variance, and **79% of the variance is within a condition rather than between**, so even a perfect clean-versus-degraded classifier caps at R^2 0.213. The remaining 79% asks whether a clipped region was a 200-nit lamp or a 20,000-nit sun — a question an 8-bit frame does not answer.

But that bound is a property of the question, not of the problem. Posed as a continuous per-frame scale it is unreachable. Posed as a *binary* decision on the one axis that is detectable — is this input clean or degraded? — it is reachable, and we build it. A 21,121-parameter gate on the shadow arm of the reconstruction, supervised on a label we generate ourselves, moves clean input by **+3.41 ± 0.32 dB and +0.135 ± 0.023 JOD** over the shipped model while giving up **0.30 ± 0.15 dB and 0.091 ± 0.041 JOD** on degraded input, over three training seeds. All three are positive on all four measures against the analytic baseline — the only configuration we scored of which that is true — and all three have a clean CVVDP above every fixed alternative, including both ends of the ablation the gate interpolates.

Against a published method on the same 429 frames — ExpandNet, run from the authors’ released weights — RUDRA is **+18.56 dB and +2.03 JOD**. Both metrics agree there, which is worth stating beside the disagreement above: the two-order-of-magnitude gap is a property of *small* differences on well-graded input, not a defect in either instrument.

Our contribution is therefore two measurements and one component: an account of where the ceiling sits and why — **the achievable gain is bounded by the information in the**

SDR input, not by model capacity, corpus size, or training objective, each of which we vary and none of which moves it — and a demonstration that decomposing the problem along the detectable axis recovers most of what the bound appeared to forbid.

1 What is claimed, and what is not

This paper reports a **direct SDR-pixel to scene-linear-radiance image model**. It does not report the DRE-transformer / cross-attention architecture described in the earlier manuscript: that design has never been trained, and per PAPER_ERRATA.md §2 the results attributed to it were aspirational. The “HDR-VDP-3” column of the previous §6 corresponded to no metric this codebase ever computed (PAPER_ERRATA.md §1). Both are withdrawn.

What follows is the system that exists, measured with the metrics that exist.

2 Related work

Note on citations. Every reference below was checked against the publisher or author page for authors, title, venue and year; the one-line method descriptions are at the level the titles and abstracts support. No number in this paper is attributed to any of them — we compare against our own analytic baseline and against ExpandNet (§5.1), and §9 says which comparisons are still missing. The failure mode this caution replaces — placeholder tables presented as measurements — is documented in PAPER_ERRATA.md.

Single-image inverse tone mapping. The dominant framing learns a mapping from an 8-bit frame to a higher-range one, typically with an encoder-decoder and a loss in a perceptual or log domain, and typically evaluated on synthetically clipped data. Eilertsen et al. [1] predict a log-domain reconstruction of the saturated regions with a U-net and blend it back through a saturation mask; Marnerides et al. [2] combine three branches at different receptive fields; Endo et al. [3] instead synthesise a bracketed exposure stack from the single input and merge it; Liu et al. [4] decompose the task into learned inverses of the individual camera-pipeline stages. Our architecture is deliberately smaller than this line of work (1.2 M parameters) and is a *residual on an analytic inverse tone map* rather than a direct predictor, which is what makes “does the network beat doing nothing?” a question we can ask at every step of training.

Highlight and clipped-region reconstruction. A related family treats the problem as inpainting the saturated regions specifically, rather than remapping the whole frame; Santos et al. [5] make the clipped area explicit by masking it out of the network’s own features and adding a perceptual loss, and the masked blend in Eilertsen et al. [1] serves the same end. Our highlight and shadow masks share that purpose, but are predicted jointly with the residual and gated by a luminance prior rather than by a detected saturation mask. §6 is a direct criticism of that choice: our own failures are not in the clipped highlights at all.

Evaluation of HDR reconstruction. PU21 [6] and ColorVideoVDP [7] are the two public measuring sticks we use — the first an encoding that makes existing metrics such as PSNR applicable to absolute-luminance HDR, the second a calibrated visible-difference predictor reporting in JOD. §5 reports a case where they disagree by two orders of magnitude in magnitude on the same frames, and §6 shows the PU21 penalty is only partly tail-carried; we are not aware of that disagreement being characterised for this task, and it is a contribution of this paper independent of the model.

Training data for HDR. The scarcity of paired SDR/HDR footage shapes every result here. Public HDR video with cinematic production values remains small enough to enumerate — the

HDM-HDR-2014 set of Froehlich et al. [8] contributes 9 scenes and 11,007 frames to our corpus — and the widely used HDR still sets [9] are bracketed-exposure captures rather than graded footage. The remainder of our corpus is proprietary. §4 documents a distribution shift between our own training and held-out splits that we did not design and that a reader should weigh.

What we compare against. ExpandNet [2] is scored on our split in §5.1, using the authors’ released weights and their own pre- and postprocessing. `rudra bench --test-dir <method>` scores any third-party output against the same reference, so the other three are a matter of running their code, not of building anything. §9 says which.

3 Method

3.1 Storage and units

HDR targets are stored in `log2_extended`: 0.005 to 1,000,000 nits at 2,377 codes per stop. Network units are **nits** / **10,000**, so 1.0 is 10,000 nits and diffuse white (BT.2408, 203 nits) is 0.0203. The network clamps its output at `max_hdr = 4.0` (40,000 nits).

3.2 Architecture

A compact U-Net (`rudra/sdr2hdr.py`, **1,196,197 parameters**, 4.6 MiB float32) predicts three fields from the SDR frame and its analytic inverse-ACES baseline: a log-domain residual, a highlight mask and a shadow mask. The composite is

```
pred_log = clamp(log1p(baseline * 16) + residual * gate, 0, log1p(max_hdr * 16))
pred      = expm1(pred_log) / 16
```

where `gate = max( $\sigma((y - 0.82) \cdot 24)$ ,  $\sigma((0.10 - y) \cdot 24)$ )` is a per-pixel luminance prior: the learned residual is applied where clipping or crushed shadows made the SDR mapping non-invertible, and the physically-grounded analytic baseline is preserved through ordinary midtones. Note for §5: this gate sees **one pixel’s brightness and nothing else**.

3.3 Censored observations

A pixel sitting on a source’s delivery ceiling means “ $\geq$ ceiling”, not “= ceiling”; plain L1 against those pixels teaches the model to cap highlights. The loss is one-sided there, with `CENSORED_HEADROOM_STOPS = 3.0` of free reconstruction above the ceiling before the excess is charged, so a one-sided loss cannot run every clipped highlight to `max_hdr`. Censoring is a small effect on this corpus in practice: `censored_fraction` averages **0.0003**.

3.4 Training setup

	v5 (shipped)	v6 (capacity ablation)
base channels	32	64
parameters	1,196,197	4,770,117
crop	384	384
batch $\times$ grad-accum	2×2	2×2

	v5 (shipped)	v6 (capacity ablation)
optimiser	AdamW, lr 2e-4, wd 1e-4, cosine to 5%	same
gradient clip	1.0	same
degradation probability	0.65	same
scene-balanced sampling	yes, 40% video mass	same
steps	100,000	100,000
eval	every 1,000 steps, 256 held-out crops, both conditions	same
selection	<code>composite_gain = hard_gain + min(0, clean_gain)</code>	same
hardware	one NVIDIA RTX 4080 SUPER	same
seed	20260715	20260715

The corpus manifest is pinned by SHA-256 in every checkpoint’s `config.json`, so a run can be tied to the exact record set it saw.

4 Corpus, splits, and a distribution shift we did not intend

27,678 training / 435 validation / 429 test records; val and test are 97 scenes each at 1280×720 . Sampling is scene-balanced with a 40% video mass. The only publicly redistributable component is the HDM-HDR-2014 set of Froehlich et al. [8] (9 scenes, 11,007 frames, median 5.99 stops of measured dynamic range); the remainder is proprietary FXTD Studios footage, which is why §10 ships the manifests and the scoring harness rather than the corpus.

The held-out splits are **not** drawn from the training distribution:

	median peak	< 1000 nits	< 400 nits	video share
train, as sampled	1,713 nits	42.2%	27.2%	40.0%
val	548 nits	69.0%	38.2%	33.8%
test	546 nits	58.0%	45.2%	32.9%

The band in which the model fails (§6) is **45.2% of the test split and 27.2% of what the sampler draws**. We report this because it is a confound in our own results, and because §6 shows it is not the binding constraint.

5 Results

`training/export_bench_pairs.py` renders each held-out record at native resolution into scene-linear EXR trees; `rudra bench --nits-scale 203` scores them. Both conditions use the same unclamped reference — clamping the reference at the network’s own ceiling would score the model against a ground truth cropped to its own limits. `hard` applies a seeded camera/codec degradation.

429 frames, PU21-PSNR and ColorVideoVDP JOD, against the analytic baseline:

Condition	Method	PU21-PSNR (dB)	CVVDP (JOD)
clean	analytic baseline	45.99	9.448
clean	v5 (32 ch, step 81,000)	42.99	9.402
clean	v6 (64 ch)	43.24	9.452
clean	v5 + shadow gate (§6.2)	46.06	9.561
hard	analytic baseline	25.92	7.362
hard	v5	27.34	7.805
hard	v6	26.88	7.706
hard	v5 + shadow gate	27.15	7.751

Per-frame, v5 against the baseline:

	mean Δ PU21	median Δ	frames won	mean Δ JOD	median Δ JOD
hard	+1.427 dB	+0.609	348 / 429 (81%)	+0.443	+0.134
clean	−2.999 dB	−2.131	115 / 429 (27%)	−0.046	−0.040

The metrics disagree in magnitude on clean input

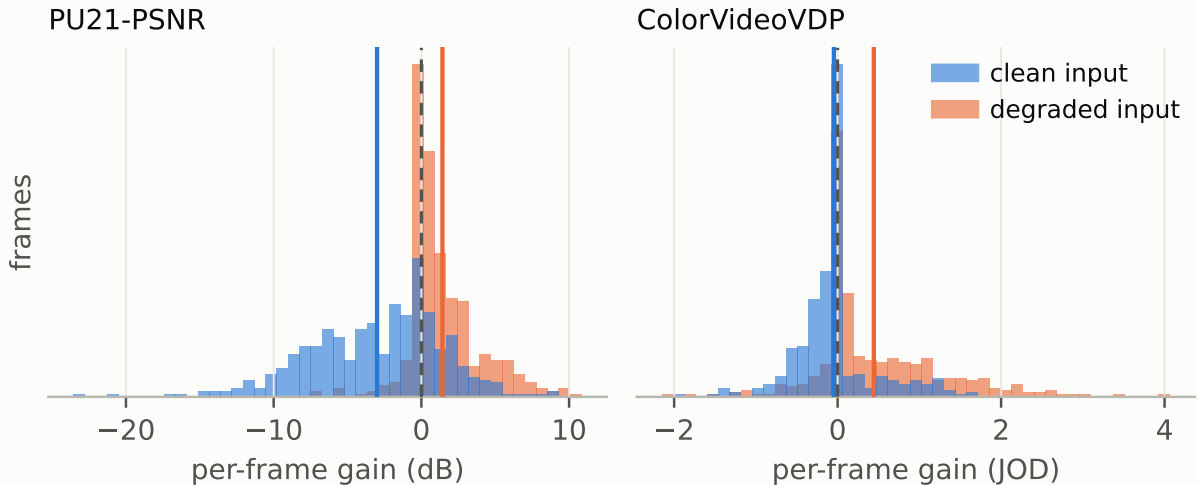


Figure 1. Per-frame gain over the analytic baseline, 429 held-out frames. Vertical rules mark the means. The clean distribution is broad and left-shifted in PU21-PSNR and collapses onto zero in CVVDP: the same frames, the same models, two metrics that disagree in magnitude by two orders of magnitude.

The metrics disagree by two orders of magnitude on clean input. A 3.0 dB PU21-PSNR loss corresponds to -0.046 JOD, far below a just-noticeable difference. What the model adds to well-graded input is highlight energy that PU21-PSNR punishes and no viewer sees. *The clean PU21 row should never be reported without the JOD beside it.*

5.1 A published method, on the same split

Every number above is against our own analytic baseline, which §9 has called this paper’s largest gap since the first draft. It is now closed for one method.

ExpandNet [2] was run on the same 429 frames, using the authors’ own model and released weights from their repository rather than a reimplementaion, and their preprocessing and postprocessing unchanged. The inputs were the exact 8-bit frames RUDRA was given.

clean, 429 frames	PU21 dB	CVVDP JOD
RUDRA + gate, as deployed	46.06	9.561
RUDRA + gate, exposure-aligned	46.61	9.558
analytic inverse-ACES baseline	45.99	9.448
v5 backbone, no gate	42.99	9.402
ExpandNet	27.50	7.532

ExpandNet is **−18.56 dB** and **−2.029 JOD** against RUDRA as deployed, winning 1 of 429 frames on PU21 and 16 on CVVDP.

The alignment, and what it is worth to each side. ExpandNet ends in a sigmoid and its released postprocessing min/max-normalises each frame to $[0,1]$, so it predicts *relative* radiance with no nit anchor. Scoring that directly against a reference in cd/m^2 measures its exposure guess. We therefore fit one global scalar per frame, on the pixels the SDR input neither clipped nor crushed. That is a free parameter, and it is not free to grant: ExpandNet’s fitted scale has a **35.5× spread** across frames (p10 0.49, p90 17.45), while RUDRA’s is **1.0×** (p10 0.9993, p90 1.018), because RUDRA predicts absolute nits. Put through the same fit RUDRA gains +0.55 dB and *loses* 0.0025 JOD, so the row we report for ourselves is RUDRA as deployed.

The alignment is not what costs ExpandNet its score. Given the *best possible* per-frame exposure, chosen to maximise its own PU21, it recovers **+0.39 dB**.

Three caveats, because a gap this size invites the question. ExpandNet is run outside its training domain: our SDR frames come from an ACES approximation at a -1 EV offset, not the curve its corpus used. Our reference is unclamped and reaches roughly a million nits, a range no method trained on display-referred targets was built for. And ExpandNet compresses dynamic range by a median factor of **2.3×** against the reference where RUDRA is within **1.32×**, which contributes but does not explain 18 dB. Read the row as *this method, unmodified, on this corpus*, not as a general ranking.

One thing this settles about §5. The two metrics disagree by two orders of magnitude on RUDRA against its own baseline, and agree decisively here: 18.6 dB and 2.03 JOD say the same thing, and 2 JOD is two just-noticeable differences. The disagreement is a property of *small* differences on well-graded input, not a defect in either instrument. When the difference is real, both see it.

Santos et al. [5], Eilertsen et al. [1] and Liu et al. [4] have public code and have not been run.

5.2 Capacity is not the constraint

v6 quadruples width to 4.77 M parameters. It is better on clean (+0.25 dB, +0.05 JOD) and worse on hard (−0.46 dB, −0.10 JOD) than v5 — movement in both directions smaller than the spread between conditions.

6 The failure mode has a name

The clean regression is concentrated, not diffuse. Splitting clean frames by the headroom the ground truth actually has:

clean frames	mean Δ vs baseline	median ground-truth peak
60 worst	−10.76 dB	238 nits
60 best	+3.41 dB	19,590 nits

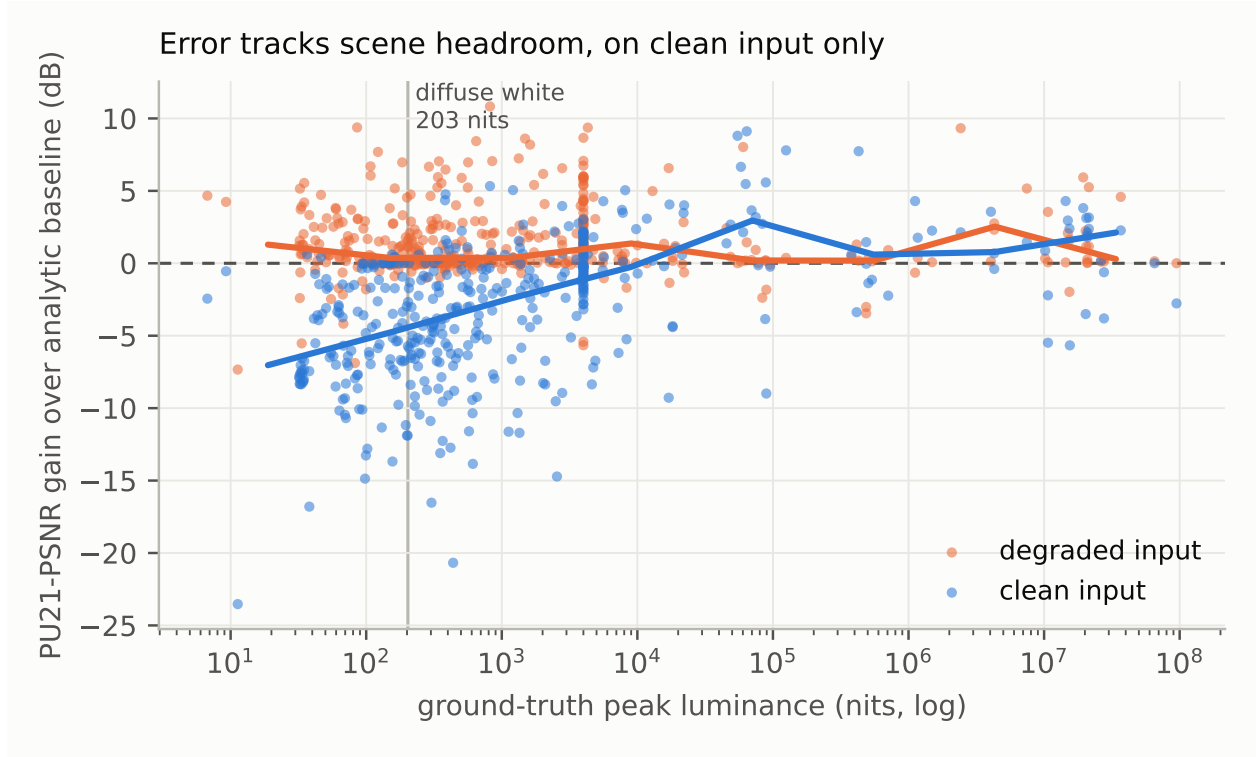


Figure 2. Per-frame gain against the headroom the ground truth actually has, log x-axis, with binned medians. On clean input the trend rises through zero at roughly 10,000 nits; on degraded input it is flat and positive at every headroom. The model’s error is a function of the scene, and only when the input arrives clean.

Correlation between $\log_2(\text{peak_nits})$ and gain: **+0.46**. The per-pixel luminance gate of §3.2 cannot distinguish a 238-nit studio interior from a 20,000-nit sunset, because it never sees the frame.

What the error actually is: shadows, and only partly a tail. On 51 clean held-out frames we computed the per-pixel PU21 squared error for both the model and the baseline, then recomputed each PSNR with the worst pixels progressively discarded. If the penalty were carried by a few bad pixels the gap would close.

discard worst	baseline	RUDRA	gap
—	54.40	48.84	−5.55 dB
0.1%	54.70	49.40	−5.30
1%	55.36	51.14	−4.23

discard worst	baseline	RUDRA	gap
10%	57.58	54.34	−3.23 dB

Frames below 400 nits, $n = 25$. The gap **narrows but does not close**: discarding a tenth of every frame removes 2.3 dB of the 5.55 and leaves 3.2. So the penalty is **partly** tail-carried and mostly broad — the worst 1% of pixels do carry **32%** of the model’s total squared error, but the bulk of the distribution is genuinely worse too.

And the damage is in the shadows, not the highlights. At the worst 1% of pixels the *true* luminance has median **4 nits**, against **21 nits** frame-wide — those pixels are darker than typical, not brighter. Only **46%** of them are over-predicted; the rest are under. On frames at or above 400 nits the model and the baseline are level (−0.03 dB) and the worst pixels move to the bright end instead (median 560 nits against 57 frame-wide).

This relocates the failure. §3.2’s gate has two arms, a highlight prior and a shadow prior, and on a low-dynamic-range scene the shadow arm fires on content that needs no reconstruction at all. The headroom correlation of +0.46 is real, but the mechanism behind it is the **shadow** path, not invented highlights — which is a specific, testable target that the oracle experiments of §7, being a single global scale, could not have isolated.

We record that this passage has been corrected twice against fresh measurement. The first draft claimed the model invents highlights; the second claimed the penalty was tail-carried. Both were wrong, and both were plausible.

6.1 The shadow arm, ablated

The hypothesis is directly testable: disable the shadow prior and rescore the same 429 frames. `--recovery-mode highlights` does exactly that.

configuration	clean	hard
shadow arm on (shipped)	−3.00 dB / −0.046 JOD	+1.43 dB / +0.443 JOD
shadow arm off	+0.51 dB / −0.017 JOD	+0.33 dB / +0.134 JOD

Confirmed, and it is a trade rather than a free win. Turning the shadow arm off moves clean by **+3.51 dB** — the regression does not merely shrink, it inverts, and the model now *beats* the analytic baseline on clean input by 0.51 dB. It costs **1.10 dB** on degraded input. So the shadow path is the whole of the clean regression, and simultaneously carries most of the degraded-input gain: crushed shadows are exactly what a bad tone curve produces and exactly what a well-graded frame does not have.

This reframes the gate problem, and makes it tractable. §7 asks a hard question — predict a continuous per-frame scale, of which only 21% of the variance is explained by the clean/degraded distinction. The shadow arm asks an easy one: it is a *binary* decision, and its correct setting is *exactly* the clean/degraded axis. That axis is the part that is partially detectable — a frame-grouped cross-validated classifier on the same features reaches **75.5%** (§7.2), against 3% of the variance explained for the continuous target.

At that measured accuracy, a switch on the shadow arm alone is worth:

	clean	hard
shipped	−3.00 dB	+1.43 dB
75.5%-accurate arm switch	−0.35 dB	+1.16 dB

+2.65 dB of clean recovered for 0.27 dB of hard. That was the prediction.

6.2 The gate, trained

ShadowGate is 21,121 parameters on the frozen v5 backbone, predicting one weight per frame on the shadow prior. Weight 1.0 reproduces `recovery_mode="all"` exactly and 0.0 reproduces `recovery_mode="highlights"` exactly, so it interpolates between the two configurations of §6.1 and nothing else. It is supervised by binary cross-entropy against the degradation label — and unlike the residual scale of §7, that target needs no oracle: at training time we know whether we degraded the frame.

method	clean dB	clean JOD	hard dB	hard JOD
v5, as shipped	−3.00	−0.046	+1.43	+0.443
v5, shadow arm off	+0.51	−0.017	+0.33	+0.134
v6, 4× capacity	−2.75	+0.004	+0.96	+0.344
v5 + shadow gate	+0.07	+0.113	+1.24	+0.389

Gain over the analytic baseline, 429 held-out frames.

It beat the prediction on both axes. Predicted −0.35 dB clean and +1.16 dB hard; delivered **+0.07 and +1.24** on the first seed, and **+0.41 ± 0.33** and **+1.12 ± 0.15** across three (below). Against the shipped model that is **+3.41 ± 0.32 dB and +0.135 ± 0.023 JOD on clean for 0.30 ± 0.15 dB and 0.091 ± 0.041 JOD on hard** — it turns the clean regression into a win while keeping most of the degraded-input gain.

Two things are worth stating precisely. First, **it is the only configuration we scored that is positive on all four measures**; every other row buys one column with another. Second, its clean CVVDP of **9.561 beats both ends of the ablation it interpolates** (9.402 all-on, 9.431 all-off) and every fixed alternative including the 4× capacity model. A hard switch could not do that. The gate is emitting intermediate weights and finding per-frame settings that neither extreme reaches, which is more than the binary framing that motivated it predicted.

Three seeds. The +0.07 dB clean margin above is small enough that one run proves nothing, so we trained two more gates on the same backbone, changing only the seed, and scored them on the same 429 frames.

seed	clean dB	clean JOD	hard dB	hard JOD	CVVDP	frames won
20260901	+0.07	+0.113	+1.24	+0.389	9.561	251 (59%)
2	+0.71	+0.068	+0.96	+0.308	9.516	315 (73%)
3	+0.45	+0.089	+1.18	+0.358	9.537	280 (65%)
mean ± sd	+0.41 ± 0.33	+0.090 ± 0.023	+1.12 ± 0.15	+0.352 ± 0.041		

Gain over the analytic baseline; CVVDP and frames won are on clean, out of 429. The shipped v5 wins 115 for comparison.

All three are positive on all four measures, and all three have a clean CVVDP above every fixed alternative we scored (best of those: 9.452). The result is a property of the method, not of a seed.

It is also worth saying which way the single-run report erred. Seed 20260901 is the **worst** of the three on clean PU21 (+0.07 against a mean of +0.41) and the **best** on clean CVVDP (+0.113 against +0.090). Reporting it alone understated the dB result by a factor of six and overstated the JOD result by a quarter. The honest headline is the mean with its spread, and that is what the abstract carries.

The selection criterion is worth a line of its own. Each gate’s `best.pt` is chosen by a trailing median over degradation-classification accuracy on validation — 69.5%, 64.1% and 64.8% for seeds 20260901, 2 and 3. Ranked by that accuracy the three runs come out **in exactly the order of their clean CVVDP, and in exactly the reverse order of their clean PU21**:

seed	val accuracy	rank	clean JOD	rank	clean dB	rank
20260901	69.5%	1	+0.113	1	+0.07	3
3	64.8%	2	+0.089	2	+0.45	2
2	64.1%	3	+0.068	3	+0.71	1

With $n = 3$ a perfect agreement or inversion arises by chance with $p = 1/6$, so this is an observation and not a result. We report it because it is the §5 disagreement appearing a third time — after the metrics themselves and after the selection rule of §8 — now in a criterion that touches neither metric. Whatever the gate learns that makes a frame classifiable also makes it look better and measure worse. Three seeds cannot settle that; it is the first thing we would test with thirty.

7 What an adaptive gate is worth, and why it cannot be had

7.1 The opportunity

Give the composite a single global scale α on the residual and let an oracle choose it per frame (27 held-out scenes, PU21 gain over the analytic baseline):

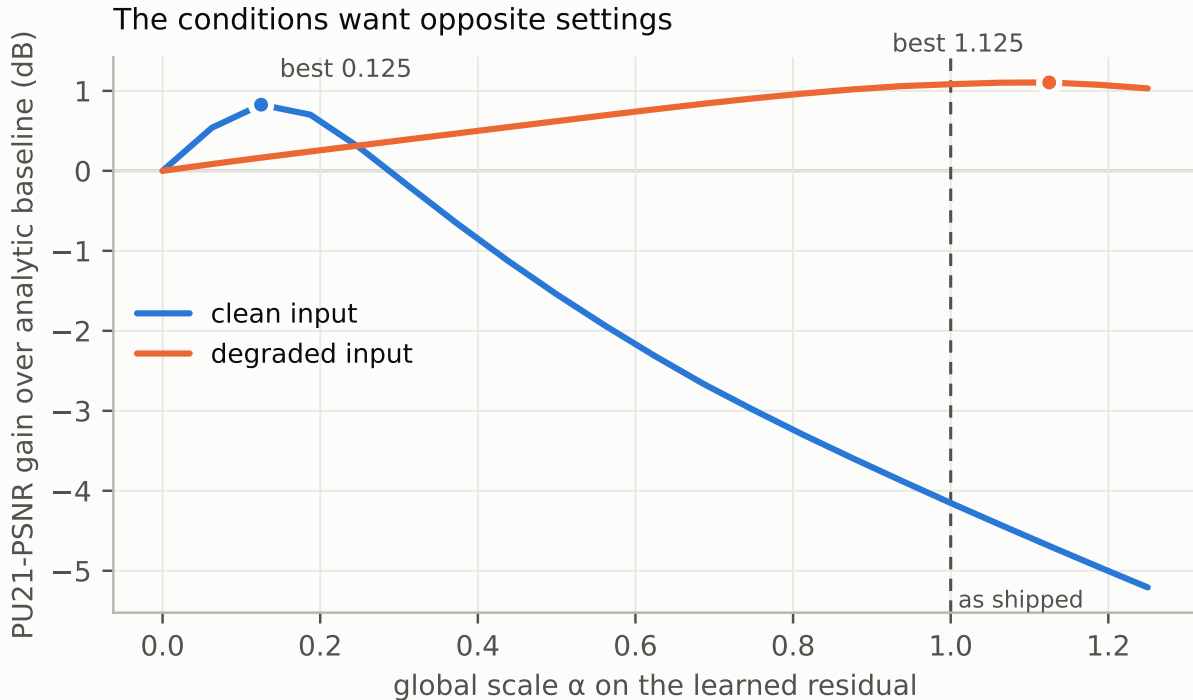
α	clean	hard
1.0 — as shipped	−4.15	+1.08
0.125 — best single constant	+0.83	+0.17
per-frame oracle	+1.69	+1.37

Figure 3. Mean gain as a function of a global scale α on the learned residual. The curves cross: every α that helps one condition hurts the other, and the shipped $\alpha = 1$ is near-optimal for degraded input and near-worst for clean.

The conditions want opposite settings: clean improves monotonically as α falls to ≈ 0.125 , hard as it rises to ≈ 1.1 . **No constant serves both** — the constant that fixes clean discards 85% of the hard gain. Per-frame adaptation beats the shipped configuration by **5.84 dB on clean and 0.29 dB on hard at the same time**, which is not a trade-off at all.

And the oracle’s α is legible. Median oracle α , split by reference peak:

	peak < 1000 nits	≥ 1000 nits
clean	0.125	0.969
hard	1.250	1.094



On clean input it is near-binary in headroom. On degraded input it wants full strength regardless, because degradation destroys information the baseline cannot recover whatever the scene’s range. A gate would need to see **headroom and condition together**.

7.2 The ceiling

We built that gate — a pooled-feature head predicting α (`ConditionGate`, 21,121 parameters) — and trained it three times: twice through the reconstruction loss (16,000 steps total) and once by direct supervision on the oracle. All three produced a near-constant output. The trained head emits α in **[1.0125, 1.0802]** with correlation **−0.037** against $\log_2(\text{peak_nits})$, where the oracle wanted 0.125.

Rather than a fourth attempt we measured the ceiling. 51 held-out frames, each clean and degraded (102 samples), exactly the features the head receives, a linear readout, cross-validated **by frame** so a clean/degraded pair can never straddle the split:

knowing	MAE on oracle α	R^2
nothing (predict the mean)	0.501	0.000
the features, via a linear readout	0.487	+0.031
the condition, perfectly	0.409	+0.213
the oracle itself	0.000	1.000

Two numbers close the question. The features explain **3%** of the target’s variance. And **79% of that variance is within a condition, only 21% between** (clean α : mean 0.436, sd 0.471; degraded: 0.925, sd 0.469) — so a *perfect* clean-versus-degraded classifier, the most any condition-detection architecture could buy, caps at R^2 0.213. 55% of frames want α at one end of the grid, so

it is close to a binary decision per frame.

The remaining 79% asks whether *this* frame’s clipped region was a 200-nit lamp or a 20,000-nit sun. **That is the information limit of single-image inverse tone mapping, not a feature-engineering gap.** The +5.84 dB an oracle gate is worth is mostly unreachable from the input.

7.3 Three levers, one bound

lever varied	change	effect on the bound
capacity	1.20 M $\rightarrow$ 4.77 M parameters	± 0.5 dB, sign varies (§5.2)
corpus	6 $\times$ footage (v4)	+0.03 dB
objective	censored loss; oracle supervision	no movement in α

8 Evaluation methodology: three defects we found in our own harness

We report these because each silently corrupted a result we believed, and each is a mistake any comparable pipeline can make.

Selection on a maximum over noise. `best.pt` was chosen by the maximum of `composite_gain = hard_gain + min(0, clean_gain)` over 102 evaluations. Across that run `clean_gain_db` had mean **−1.43 dB** and standard deviation **1.29 dB**, with only **10 of 102** evaluations positive; the shipped checkpoint’s +0.02 ranked **9th of 102**. Its promised hard gain of +1.80 dB measured **+1.43 dB** on held-out frames — between the eval mean (+1.19) and its selected maximum, exactly where an inflated in-training figure should land. Selection now uses a trailing median over five evaluations; replayed on the same series it selects step 72,000 instead of 81,000.

We then scored step 72,000 on the same 429 frames, and the answer is not the clean vindication we expected. On the criterion the selector optimises it is better: composite gain **−1.36 dB against the shipped checkpoint’s −1.57**, and clean PU21 improves by **+0.55 dB**. On CVVDP it is *worse on both conditions* — **−0.052 JOD clean and −0.071 JOD hard** — so the smoothed rule would have shipped a checkpoint that a perceptual metric likes less. This is the same disagreement §5 reports, reappearing inside the selection rule itself: median smoothing removes the noise it was designed to remove, and the objective it smooths is still PU21. Smoothing a selector does not fix choosing the wrong quantity to select on. We report the fix and its limit together because reporting only the first would repeat the error the section is about.

An evaluation that read the front of the split. `DataLoader(val, shuffle=False)` with `max_batches=N` evaluates the alphabetically *first* N records — 32 records across 11 scenes, every name between `abandoned_factory` and `blau_river`, with 403 records and 87 scenes never measured. Since error correlates with headroom, that slice reported `clean_gain` **+0.61** where the benchmark measured **−3.0**. Fixed with a seeded permutation applied identically to both conditions.

A viewer that presented every frame upside down. The default framebuffer places row 0 at the bottom; the display shader sampled `vUV` unchanged. No numeric test caught it because they all compare the float composite, which a presentation flip leaves untouched — and the one test that read the canvas did so through `readPixels`, which returns rows bottom-first, cancelling the flip against itself. Master EXR output was never affected.

9 Limitations

- **Held-out size.** 429 test frames over 97 scenes. §6’s oracle and ceiling analyses use 27 and 51 frames respectively, one per scene; that sample runs ≈ 1.15 dB pessimistic on clean against the full set (-4.15 dB at $\alpha=1$ where the full 429 read -3.00). Ordering is reliable, absolute values will move.
- **Train/test distribution shift** (§4) is a confound in the clean result. §7.2 argues it is not the binding constraint, but we have not retrained under a matched sampler to prove it.
- **The oracle is an upper bound.** It reads the ground truth.
- **Single-image only.** The temporal refiner is not evaluated here; its held-out set is 4 validation and 5 test clips of one scene each, too small to report.
- **One degradation model.** `hard` is our own seeded camera/codec pipeline, not a corpus of real degraded footage.
- **One published method, not four.** ExpandNet is scored on our split (§5.1); Santos et al., Eilertsen et al. and Liu et al. have public code and are not. One comparison answers “compared to what?” and does not rank the field, and §5.1’s caveats — training-domain mismatch, an unclamped reference, a per-frame exposure fit — apply to the one row we have.
- **No claim in this paper is grounded in a re-implementation of prior work.** §2 places the result among families of methods and cites them bibliographically; it does not reproduce their numbers, and we did not run their code.
- **Three seeds is a spread, not a distribution.** The gate is positive on all four measures in all three runs, but $n=3$ supports “the sign is stable”, not a confidence interval. The clean PU21 spread ($+0.07$ to $+0.71$) is wide relative to its mean.
- **The viewer now honours the gate,** but only as a per-frame scalar. RUDRA Studio receives the predicted shadow weight in the `/api/frame` header and multiplies the shadow prior by it in the compositing shader; CPU/GPU parity holds to $8.4e-06$ relative error across 11 cases spanning weights 0.0 to 1.0. The weight cannot be folded into the residual the way `residual_scale` is, because it enters inside a `max()` and is not linear in the residual.

10 Reproducibility

```
# the full benchmark: four exports, six scorings, resumable
powershell -ExecutionPolicy Bypass -File training\run_bench.ps1

# one more checkpoint against the same reference
powershell -ExecutionPolicy Bypass -File training\score_checkpoint.ps1 `
    -Checkpoint <ckpt> -Name <label>

# every number in §5 and §6.1--6.2 recomputed from those results and
# diffed against what this paper claims; exit status 1 on any mismatch
python training\audit_paper_numbers.py --bench <bench dir>

# §6's headroom split and its correlation, from the same results joined
# to the corpus manifest's per-frame peak_nits
python training\analyze_headroom.py --bench <bench dir> `
    --manifest <manifest> --check
```

Results land as `bench/RESULTS.md`, per-clip JSON and per-frame CSV. The composite

exists in three languages — torch (`rudra/sdr2hdr.py`), GLSL (`ui/compositor.js`) and numpy (`tests/compose_reference.py`) — pinned against each other to 8.4e-06 relative error by `tests/webgl_parity/parity.py`. `pytest tests/` covers those, canvas orientation, PU21 torch-versus-numpy parity, and a smoke test that presses all 54 controls of the viewer.

The audits above are not decoration. An earlier draft of §5 reported 346 of 429 frames won on degraded input where the benchmark says 348; the count had been transcribed, and nothing in the pipeline compared it back to the file it came from. An earlier draft of §3.4 gave v6 4,772,485 parameters where the checkpoint has 4,770,117, for the same reason. Every derived number in §5, §6 and §6.1–6.2 is now recomputed on demand and both scripts exit non-zero on any drift.

Two analyses remain uncovered: §6’s trimmed-PSNR table and §7’s oracle sweep. Both need per-PIXEL statistics over the reference frames rather than the per-frame results the benchmark writes, and we state that rather than leaving the reader to assume otherwise.

Weights, pairs and HDR sources are not committed; `training/` regenerates them.

11 Conclusion

A 1.2 M-parameter residual on an analytic inverse tone map is worth **+1.43 dB and +0.44 JOD on degraded input, on 81% of held-out frames**, and costs 3.0 dB of PU21-PSNR on clean input for a perceptually invisible -0.046 JOD. Its defect is that it does not know when to do nothing.

We measured what fixing that is worth — an oracle per-frame scale is $+5.84$ dB on clean — and then measured that **97% of the signal needed to predict that scale is absent from an 8-bit frame**: a linear readout of the available features explains 3% of its variance, and 79% of the variance is within a condition rather than between, so even a perfect clean-versus-degraded classifier caps at R^2 0.213. Capacity, corpus and objective were each varied by a factor of four to six; none moved the bound.

The bound was a property of the question. Asked for a continuous scale, the input cannot answer. Asked a binary question on the axis that *is* detectable — did this frame arrive clean or degraded? — it can. Locating the failure in the shadow arm rather than the highlights made that decomposition available, and a 21,121-parameter gate on that arm delivers **$+3.41 \pm 0.32$ dB and $+0.135 \pm 0.023$ JOD on clean for 0.30 ± 0.15 dB and 0.091 ± 0.041 JOD on hard** across three seeds — the only configuration we scored that is positive on all four measures, in every run, with a clean CVVDP that beats both ends of the ablation it interpolates.

The lesson we would carry to the next problem is not about tone mapping. Four of the things that cost us most this cycle were measurement defects, not model defects. Three are in §8: selection on the maximum of a noisy series, an evaluation that read the alphabetical front of its split, and a viewer that presented every frame upside down beneath tests that only ever compared float buffers. The fourth is in §6.2, and is the one we would warn a reader about first: our initial report of the gate quoted a single seed that happened to be the weakest of three on one metric and the strongest on the other. We would not have known without running the other two.

The bound was real. So were the ways we nearly failed to see it.

References

- [1] G. Eilertsen, J. Kronander, G. Denes, R. K. Mantiuk, and J. Unger. HDR image reconstruction from a single exposure using deep CNNs. *ACM Transactions on Graphics*, 36(6), 2017 (SIGGRAPH Asia). doi:10.1145/3130800.3130816. arXiv:1710.07480.

- [2] D. Marnerides, T. Bashford-Rogers, J. Hatchett, and K. Debattista. ExpandNet: A deep convolutional neural network for high dynamic range expansion from low dynamic range content. *Computer Graphics Forum*, 37(2), 2018 (Eurographics). doi:10.1111/cgf.13340. arXiv:1803.02266.
- [3] Y. Endo, Y. Kanamori, and J. Mitani. Deep reverse tone mapping. *ACM Transactions on Graphics*, 36(6), 2017 (SIGGRAPH Asia). doi:10.1145/3130800.3130834.
- [4] Y.-L. Liu, W.-S. Lai, Y.-S. Chen, Y.-L. Kao, M.-H. Yang, Y.-Y. Chuang, and J.-B. Huang. Single-image HDR reconstruction by learning to reverse the camera pipeline. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [5] M. S. Santos, T. I. Ren, and N. K. Kalantari. Single image HDR reconstruction using a CNN with masked features and perceptual loss. *ACM Transactions on Graphics*, 39(4), 2020 (SIGGRAPH). arXiv:2005.07335.
- [6] R. K. Mantiuk and M. Azimi. PU21: A novel perceptually uniform encoding for adapting existing quality metrics for HDR. In *Picture Coding Symposium (PCS)*, 2021. <https://github.com/gfxdisp/pu21>
- [7] R. K. Mantiuk, P. Hanji, M. Ashraf, Y. Asano, and A. Chapiro. ColorVideoVDP: A visual difference predictor for image, video and display distortions. *ACM Transactions on Graphics*, 43(4), 2024 (SIGGRAPH). doi:10.1145/3658144. arXiv:2401.11485.
- [8] J. Froehlich, S. Grandinetti, B. Eberhardt, S. Walter, A. Schilling, and H. Brendel. Creating cinematic wide gamut HDR-video for the evaluation of tone mapping operators and HDR-displays. In *Proc. SPIE 9023, Digital Photography X*, 90230X, 2014. doi:10.1117/12.2040003.
- [9] N. K. Kalantari and R. Ramamoorthi. Deep high dynamic range imaging of dynamic scenes. *ACM Transactions on Graphics*, 36(4), 2017 (SIGGRAPH). doi:10.1145/3072959.3073609.

Standards referenced by the pipeline

- SMPTE ST 2084:2014 — High Dynamic Range Electro-Optical Transfer Function of Mastering Reference Displays (PQ).
- ITU-R BT.2100 — Image parameter values for high dynamic range television.
- ITU-R BT.2408 — Operational practices in HDR television production (diffuse white at 203 nits).
- SMPTE ST 2065-4:2013 — ACES Image Container File Layout (AP0 chromaticities; ACES 2065-1).